

(a) SGLang / HiCache

turn 4 of session S

R_{10}

Prefill instance (GPU)

KV pool 82% full — evicting

resident KV_8 KV_9

needs KV_1 KV_2 KV_3 **evicted**

PCIe / pinned host 26 GB/s

Decode instance (GPU)

KV pool 12% used — 88% idle

never a candidate

Host DRAM (L2, 17.6 GB reserved) + SSD (L3)

KV_1 KV_2 KV_3

statically reserved as $\text{GPU-KV} \times \text{hcache-ratio}$

- 1 R_{10} arrives; radix lookup misses
- 2 $KV_{1..3}$ were evicted to host / SSD
- 3 fetch over PCIe, or recompute if L2 evicted too
- 4 the idle decode GPU is never a candidate

(b) Ours: GPU-first placement

turn 4 of session S

R_{10}

Prefill instance (GPU)

KV pool 82% full — evicting

resident KV_8 KV_9

needs KV_1 KV_2 KV_3

NVLink 27–53 GB/s

Decode instance (GPU)

KV pool 12% used — 88% idle

park pool in idle HBM

parked KV_1 KV_2 KV_3

Host DRAM — overflow only (priority 3)

reached only when no GPU slab fits

- 1 R_{10} arrives; radix lookup misses
- 2 park index finds $KV_{1..3}$ on the idle decode GPU
- 3 peer-GPU copy over NVLink into the local radix
- 4 prefill only the new tokens; re-park the extended prefix